

CHSH Lab

One-page application summary grounded in repo evidence on March 27, 2026.

What It Is

- Static website that presents an interactive rebuttal to Bell inequality violation claims without entanglement.
- Ships two public pages: **index.html** for the interactive narrative and **paper.html** for the formal paper artifact.

Who It Is For

- Readers evaluating low-efficiency Bell-test claims, including technically literate researchers, students, and reviewers. Inferred from the rebuttal framing in **README.md**, **index.html**, and **paper.html**.

What It Does

- Explains the Wang et al. (2025) claim and why low detection efficiency changes the meaning of an observed CHSH score.
- Shows the classical bound, Tsirelson bound, and post-selection artifact region with browser-side figures and demos.
- Lets readers manipulate angle settings, detection efficiency, and selection bias through client-side controls.
- Displays theorem statements, proof sketches, and a print-friendly paper view with KaTeX-rendered math.
- Publishes generated figure assets under **assets/figures/** and keeps the shipped surface static.
- Adds a live dashboard and share snapshot module that summarize the current demo state without any backend services.

How It Works

- Frontend: plain HTML, CSS, and ES modules under **css/** and **js/**; interaction stays in the browser.
- Content model: **index.html** carries the narrative flow; **paper.html** is the canonical paper artifact.
- Figures: **scripts/generate_figures.py** regenerates image assets into **assets/figures/** using the Python scientific stack.
- Deployment: **build.sh** creates a Vercel Build Output API bundle in **.vercel/output/** with clean URL redirects and static asset copy rules.

How To Run

- Preview locally with **python -m http.server 4317** from the repo root.
- Open **http://127.0.0.1:4317/index.html** for the interactive narrative.
- Open **http://127.0.0.1:4317/paper.html** for the formal paper and print view.
- Build the deployable static output with **bash build.sh**; output is written to **.vercel/output/**.